Rogue Merchant: A Dockerized Fantasy Bazaar

Buy, sell, and master the art of fantasy market trading — powered by Docker containers

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The Deployment Nightmare

Python Version Hell

3.9 on dev, 3.11 on prod — broken dependencies everywhere

MySQL Setup

Manual installations, schema conflicts, permission nightmares

Nginx Routing

Host-level configuration, port conflicts, SSL headaches

"It works on my machine" — the most dangerous phrase in software

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Enter Docker: The Container Revolution

Package Everything

Code, runtime, libraries, config — all bundled together

Isolated & Secure

No more conflicting dependencies or system pollution

Deploy Anywhere

Same container works on laptop, server, or cloud

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Three-Tier Architecture

Nginx Frontend

Serves static assets, reverse proxies API requests

Flask Backend

REST API handles game logic, pricing, transactions

MySQL Database

Persistent storage for prices, inventory, portfolio

Each tier runs in its own container, communicating securely over Docker networks

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Multi-Stage Builds: Smart Optimization

01

Builder Stage

Installs Python, compiles dependencies into virtual environment

02

Runner Stage

Copies only clean venv — no build tools, no bloat

03

Result

Smaller, more secure image (80% size reduction!)

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Docker Networks: Secure Communication

Isolated Bridge Network

Containers communicate over `rogue-net` — invisible to outside world

Internal DNS Magic

Nginx routes to `http://backend:5000` automatically

Port Security

Flask never exposes 5000 to host — only Nginx port 8080 is public

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Docker Volumes: Data Persistence

The Problem

Containers are ephemeral — restart means losing all player data

The Solution

Map `db_data` volume to MySQL data directory

The Result

Gold, inventory, and P&L survive crashes, restarts, updates

Real demo: Stop MySQL container, restart — your fantasy fortune stays intact!

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Live Demo: Launch & Play

1

Build & Start

```
docker compose up -d --build
```

2

Open Game

http://localhost:8080

3

Play & Profit

Buy low, sell high across 10 market rounds

4

Crash Test

```
docker compose down -v
```

Volume wipes database — fresh start

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API Endpoints & Game Logic

Game State

`/api/new-game` — 10 rounds of hidden prices seeded server-side

Market Data

`/api/market` — current prices + next-round hints

Portfolio

`/api/portfolio` — gold, holdings, net worth

Trading

`/api/buy` and `/api/sell` — update inventory

History

`/api/price-history` — sparklines for price trends

Scoring

≥1200 gold = 👑 Legendary

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Why Docker Compose Wins

1

Single Command

Up and running instantly

0

Manual Setup

No installations, no config

100%

Consistency

Works everywhere identically

Core Benefits

- Multi-stage builds for optimization
- Isolated networks for security
- Persistent volumes for state
- Version-controlled configuration

Questions? Let's discuss!

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